# Signal Processing Fundamentals

EE 471C / EE 381K-17 Wireless Communication Lab

## Review from the previous lecture

- Random processes are a generalization of random variables
- Random processes are widely used in communication systems
- Wide sense stationary random processes are special
  - Constant mean
  - Covariance that is only a function of the time difference
- Ergodicity
  - Sample averages converge to ensemble averages

# Preview of today's lecture

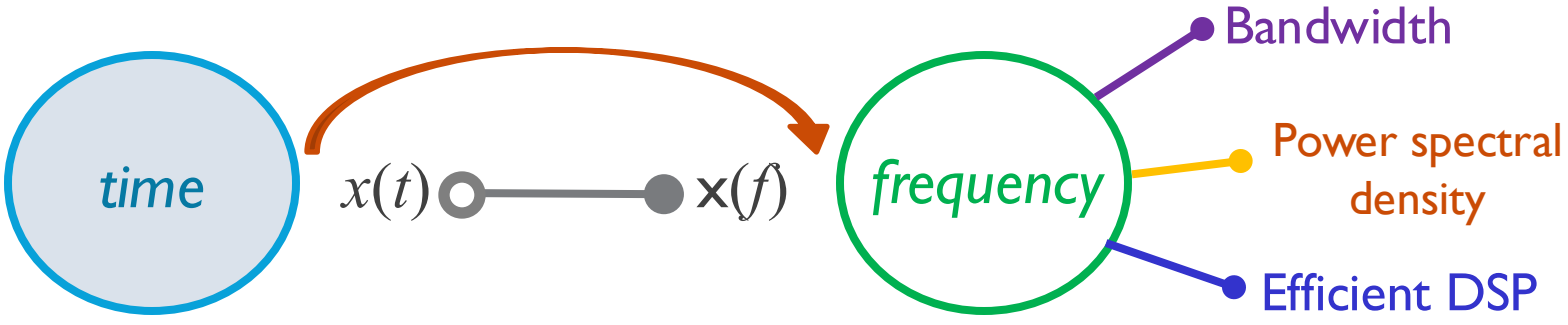
- Fourier transforms
  - Define and compute various Fourier transforms
- Band limited signals in communications
  - Explain the connection between communication and DSP
- Sampling
  - Determine the Nyquist sampling frequency for a given signal
  - Explain the theory behind sampling and reconstruction
  - Demonstrate aliasing

# Relevant Fourier Transforms

Learning objective:

- Define and compute various Fourier transforms

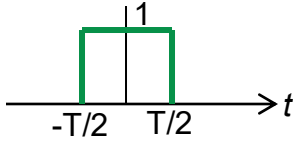
# Fourier transforms




- Several Fourier transforms arise in this course:
  - Continuous-time Fourier transform (CTFT)
  - Continuous-time Fourier series (CTFS)
  - Discrete-time Fourier transform (DTFT)
  - Discrete Fourier Transform (DFT)

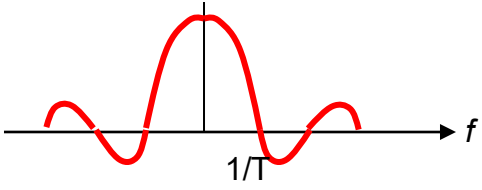


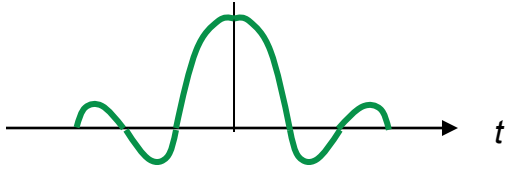
# Continuous-time Fourier transform (CTFT)



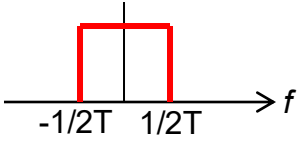

$x(t) \longleftrightarrow x(f)$

$\text{rect}(t/T) \quad \leftrightarrow \quad T\text{sinc}(fT)$




$\text{sinc}(t/T) \quad \leftrightarrow \quad T\text{rect}(fT)$




- Relates a continuous time function $x(t)$ and its Fourier transform

$$\text{Analysis} \quad x(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi f t} dt$$

$$\text{Synthesis} \quad x(t) = \int_{-\infty}^{\infty} x(f) e^{j2\pi f t} df .$$

| Property | Aperiodic signal | Fourier transform |
|---|---|---|
| | $x(t)$<br>$y(t)$ | $\mathrm{x}(f)$<br>$\mathrm{y}(f)$ |
| Linearity | $ax(t) + by(t)$ | $a\mathrm{x}(f) + b\mathrm{y}(f)$ |
| Time shifting | $x(t - t_0)$ | $e^{-\mathrm{j}2\pi f t_0}\mathrm{x}(f)$ |
| Frequency shifting | $e^{\mathrm{j}2\pi f_0 t}x(t)$ | $\mathrm{x}(f - f_0)$ |
| Conjugation | $x^*(t)$ | $\mathrm{x}^*(-f)$ |
| Time reversal | $x(-t)$ | $\mathrm{x}(-f)$ |
| Time scaling | $x(at)$ | $\frac{1}{\lvert a\rvert}X\left(\frac{f}{a}\right)$ |
| Convolution | $x(t) * y(t) = \int_{-\infty}^{\infty} x(\tau)y(t - \tau)\mathrm{d}\tau$ | $\mathrm{x}(f)\mathrm{y}(f)$ |
| Autocorrelation | $x(t) * x^*(-t)$ | $\lvert\mathrm{x}(f)\rvert^2$ |
| Multiplication | $x(t)y(t)$ | $\mathrm{x}(f) * \mathrm{y}(f) = \int_{-\infty}^{\infty} X(\nu)Y(f - \nu)\mathrm{d}\nu$ |
| Differentiation in time | $\frac{d^n x(t)}{dt^n}$ | $(\mathrm{j}2\pi f)^n\,\mathrm{x}(f)$ |
| Integration | $\int_{-\infty}^{t} x(\tau)\mathrm{d}\tau$ | $\frac{1}{\mathrm{j}2\pi f}\mathrm{x}(f) + \frac{1}{2}X(0)\delta(f)$ |
| Differentiation in frequency | $t^n x(t)$ | $\left(\frac{\mathrm{j}}{2\pi}\right)^n \frac{\mathrm{d}^n X(f)}{\mathrm{d}f^n}$ |
| Modulation (1) | $x(t)e^{\mathrm{j}2\pi f_0 t}$ | $\mathrm{x}(f - f_0)$ |
| Modulation (2) | $x(t)\cos(2\pi f_0 t)$ | $\frac{1}{2}\left[\mathrm{x}(f - f_0) + X(f + f_0)\right]$ |
| Modulation (3) | $x(t)\sin(2\pi f_0 t)$ | $\frac{1}{\mathrm{j}2}\left[\mathrm{x}(f - f_0) - X(f + f_0)\right]$ |

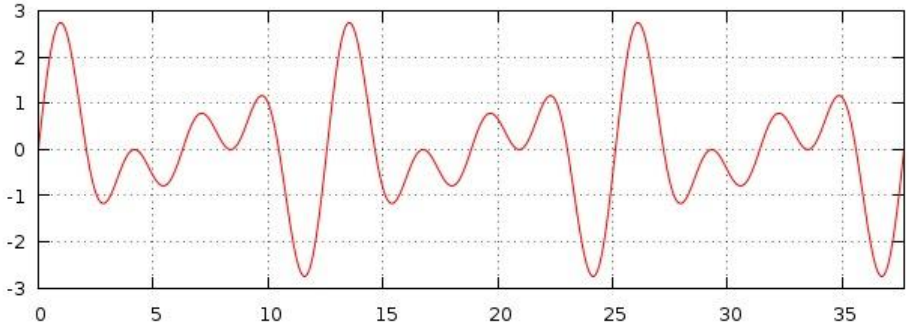
| | | |
|---|---|---|
| Conjugate symmetry for real signals | $x(t)$ is real | $\begin{cases} X(f) = X^*(-f) \\ \mathrm{Re}\{X(f)\} = \mathrm{Re}\{X(-f)\} \\ \mathrm{Im}\{X(f)\} = -\mathrm{Im}\{X(-f)\} \\ \lvert X(f)\rvert = \lvert X(-f)\rvert \\ \measuredangle X(f) = -\measuredangle X(-f) \end{cases}$ |
| Symmetry for real and even signals | $x(t)$ real and even | $X(f)$ real and even |
| Symmetry for real and odd signals | $x(t)$ real and odd | $X(f)$ purely imaginary and odd |
| Even-odd decomposition for real Signals | $x_e(t) = \mathrm{Ev}\{x(t)\}$ $[x(t)\text{real}]$<br>$x_o(t) = \mathrm{Od}\{x(t)\}$ $[x(t)\text{real}]$ | $\mathrm{Re}\{X(f)\}$<br>$j\mathrm{Im}\{X(f)\}$ |
| Duality | $x(t) \longleftrightarrow X(f)$ | $X(t) \longleftrightarrow x(-f)$ |
| Parseval's theorem<br><br>($x(t)$ and $y(t)$ real) | $\int_{-\infty}^{\infty} \lvert x(t)\rvert^2\,dt = \int_{-\infty}^{\infty} \lvert X(f)\rvert^2\,df$<br>$\int_{-\infty}^{\infty} x(t)y^*(t)dt = \int_{-\infty}^{\infty} X(f)Y^*(f)df$<br>$\int_{-\infty}^{\infty} x(t)y(-t)dt = \int_{-\infty}^{\infty} X(f)Y(f)df$ | |

| Function name | Time-domain signal $x(t)$ | Frequency-domain signal $\mathsf{x}(f)$ |
|---|---|---|
| Impulse | $\delta(t)$ | $1$ |
| DC | $1$ | $\delta(f)$ |
| Complex exponential | $e^{\mathrm{j}2\pi f_0 t}$ | $\delta(f-f_0)$ |
| Cosine | $\cos(2\pi f_0 t+\theta)$ | $\frac{1}{2}\left[e^{\mathrm{j}\theta}\delta(f-f_0)+e^{-\mathrm{j}\theta}\delta(f+f_0)\right]$ |
| Sine | $\sin(2\pi f_0 t+\theta)$ | $\frac{1}{2j}\left[e^{\mathrm{j}\theta}\delta(f-f_0)-e^{-\mathrm{j}\theta}\delta(f+f_0)\right]$ |
| Unit step | $u(t)=\begin{cases}1 & t\geqslant 0\\ 0 & t<0\end{cases}$ | $\frac{1}{2}\delta(f)+\frac{1}{\mathrm{j}2\pi f}$ |
| Sign | $\operatorname{sgn}(t)=\begin{cases}1 & t\geqslant 0\\ -1 & t<0\end{cases}$ | $\frac{1}{\mathrm{j}\pi f}$ |
| Impulse train | $\mathrm{III}(t/T)=\sum_{k=-\infty}^{\infty}\delta(t-kT)$ | $\frac{1}{T}\sum_{n=-\infty}^{\infty}\delta(f-\frac{n}{T})$ |
| Fourier series | $\sum_{k=-\infty}^{\infty}a_k e^{\mathrm{j}2\pi f_0 kt}$ where $a_k=\frac{1}{T}\int_T x(t)e^{-\mathrm{j}2\pi f_0 kt}\mathrm{d}t$ | $\sum_{n=-\infty}^{\infty}a_n\delta(f-nf_0)$ |
| Rectangle pulse | $\operatorname{rect}\left(\frac{t}{T}\right)=\begin{cases}1 & \lvert t\rvert\leqslant\frac{T}{2}\\ 0 & \text{elsewhere}\end{cases}$ | $T\operatorname{sinc}(fT)=\frac{\sin(\pi fT)}{\pi f}$ |
| Triangle pulse | $\Lambda\left(\frac{t}{W}\right)=\begin{cases}1-\frac{\lvert t\rvert}{W} & \lvert t\rvert\leqslant W\\ 0 & \text{elsewhere}\end{cases}$ | $W\operatorname{sinc}^2(fW)$ |
| Sinc pulse | $\operatorname{sinc}(Wt)=\frac{\sin(\pi Wt)}{\pi Wt}$ | $\frac{1}{W}\operatorname{rect}\left(\frac{f}{W}\right)$ |
| Sinc$^2$ pulse | $\operatorname{sinc}^2(Wt)$ | $\frac{1}{W}\Lambda\left(\frac{f}{W}\right)$ |
| Exponential pulse | $e^{-a\lvert t\rvert}$ with $a>0$ | $\frac{2a}{a^2+(2\pi f)^2}$ |
| Decaying exponential | $e^{-at}u(t)$ with $\operatorname{Re}\{a\}>0$ | $\frac{1}{a+\mathrm{j}2\pi f}$ |
| | $te^{-at}u(t)$ with $\operatorname{Re}\{a\}>0$ | $\frac{1}{(a+\mathrm{j}2\pi f)^2}$ |
| | $\frac{t^{n-1}}{(n-1)!}e^{-at}u(t)$ with $\operatorname{Re}\{a\}>0$ | $\frac{1}{(a+\mathrm{j}2\pi f)^n}$ |
| Linear decaying | $\frac{1}{t}$ | $-\mathrm{j}\pi\operatorname{sgn}(f)$ |

# Continuous-time Fourier series (CTFS) (1)

- Consider a periodic signal of period $T$

$$x(t) = x(t+T)$$

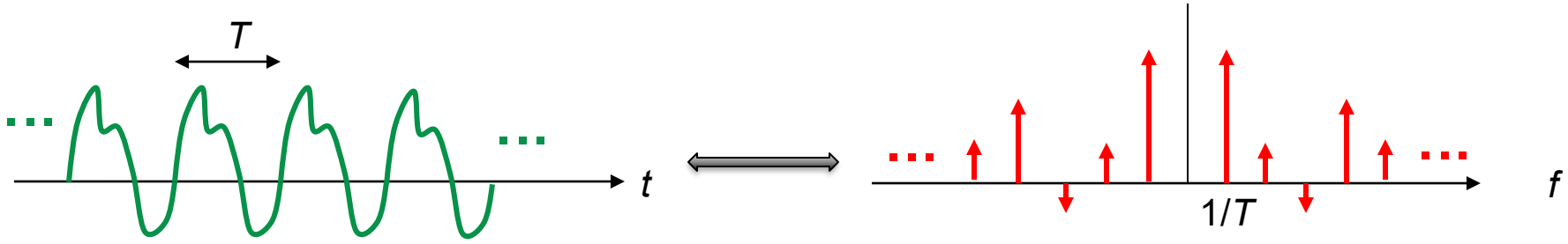



- The continuous-time Fourier series (CTFS) of a periodic signal $x(t)$ is

$$\text{Analysis} \quad x[n] = \frac{1}{T}\int_0^T x(t)e^{-j\frac{2\pi}{T}nt}dt$$

$$\text{Synthesis} \quad x(t) = \sum_{n=-\infty}^{\infty} x[n]e^{j\frac{2\pi}{T}nt}.$$

# Continuous-time Fourier series (CTFS) (2)

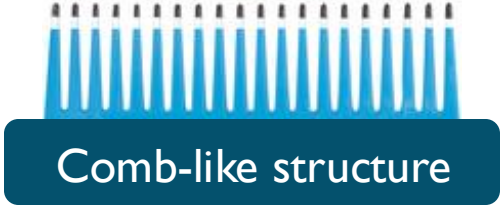



- *Define* the Fourier transform of a periodic signal

**Fourier series coefficients**

$$\text{Analysis} \quad x(f) = \sum_n x[n]\delta\left(f - \frac{1}{T}n\right)$$

$$\text{Synthesis} \quad x(t) = \int_f x(f)e^{j2\pi f t}df.$$



# Discrete-time Fourier transform (DTFT)

- For a well behaved discrete time signal

$$\text{Analysis} \quad x(e^{j2\pi f}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j2\pi f n}$$

$$\text{Synthesis} \quad x[n] = \int_{-1/2}^{1/2} x(e^{j2\pi f n}) e^{j2\pi f n} df .$$

- The DTFT is also often written in radians using $w = 2\pi f$
- For the DTFT, $f$ is only unique on any finite interval of unit length
  - Typically $f \in (-1/2, 1/2]$ $f \in [0, 1)$
  - Low frequencies are around 0 or 1, high are around -1/2 or 1/2

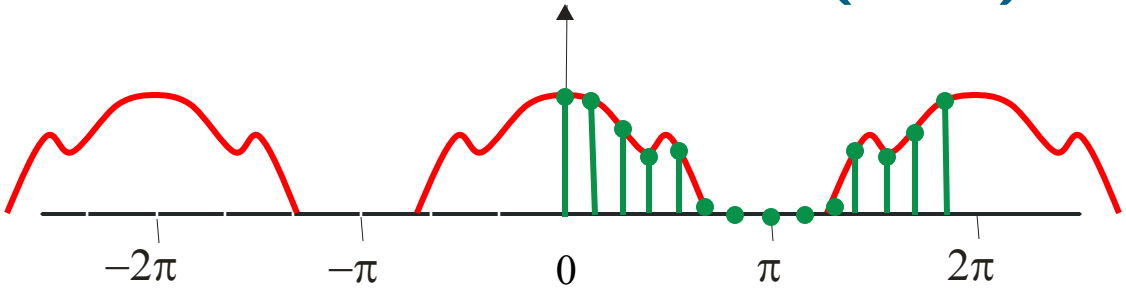
| Property | Sequence | Fourier transform |
|---|---|---|
| | $x[n]$ | $\mathsf{x}(e^{\mathrm{j}2\pi f})$ |
| | $y[n]$ | $\mathsf{y}(e^{\mathrm{j}2\pi f})$ |
| Linearity | $ax[n] + by[n]$ | $a\mathsf{x}(e^{\mathrm{j}2\pi f}) + b\mathsf{y}(e^{\mathrm{j}2\pi f})$ |
| Time Shifting | $x[n - n_0]$ | $e^{-\mathrm{j}2\pi f n_0}\mathsf{x}(e^{\mathrm{j}2\pi f})$ |
| Frequency Shifting | $e^{\mathrm{j}2\pi f_0 n}x[n]$ | $\mathsf{x}(e^{\mathrm{j}2\pi (f-f_0)})$ |
| Conjugation | $x^*[n]$ | $\mathsf{x}^*(e^{-\mathrm{j}2\pi f})$ |
| Time Reversal | $x[-n]$ | $\mathsf{x}(e^{-\mathrm{j}2\pi f})$ |
| Conjugate Symmetry | $x[n]$ real | $\mathsf{x}(e^{\mathrm{j}2\pi f}) = \mathsf{x}^*(e^{-\mathrm{j}2\pi f})$ |
| Even Symmetry | $x[n] = x[-n]$ | $\mathsf{x}(e^{\mathrm{j}2\pi f}) = \mathsf{x}(e^{-\mathrm{j}2\pi f})$ |
| Odd Symmetry | $x[n] = -x[-n]$ | $\mathsf{x}(e^{\mathrm{j}2\pi f}) = -\mathsf{x}(e^{-\mathrm{j}2\pi f})$ |
| Convolution | $x[n] * y[n] = \sum_{m=-\infty}^{\infty} x[m]y[n-m]$ | $\mathsf{x}(e^{\mathrm{j}2\pi f})\mathsf{y}(e^{\mathrm{j}2\pi f})$ |
| Autocorrelation | $x[n] * x^*[-n]$ | $\lvert\mathsf{x}(e^{\mathrm{j}2\pi f})\rvert^2$ |
| Multiplication | $x[n]y[n]$ | $\int_{-1/2}^{1/2} \mathsf{x}(e^{\mathrm{j}2\pi\theta})Y(e^{\mathrm{j}2\pi(f-\theta)})\mathrm{d}\theta$ |
| Multiplication by $n$ | $nx[n]$ | $\frac{1}{-\mathrm{j}2\pi}\frac{d}{df}\mathsf{x}(e^{\mathrm{j}2\pi f})$ |
| Sum | $\sum_{n=-\infty}^{\infty} x[n]$ | $\mathsf{x}(e^{\mathrm{j}2\pi 0})$ |
| Value at Origin | $x[0]$ | $\int_{-1/2}^{1/2} \mathsf{x}(e^{\mathrm{j}2\pi f})\mathrm{d}f$ |
| Modulation (1) | $x[n]e^{\mathrm{j}2\pi f_0 n}$ | $\mathsf{x}(e^{\mathrm{j}2\pi (f-f_0)})$ |
| Modulation (2) | $x[n]\cos(2\pi f_0 n)$ | $\frac{1}{2}\left[\mathsf{x}(e^{\mathrm{j}2\pi (f-f_0)}) + \mathsf{x}(e^{\mathrm{j}2\pi (f+f_0)})\right]$ |
| Modulation (3) | $x[n]\sin(2\pi f_0 n)$ | $\frac{1}{j2}\left[\mathsf{x}(e^{\mathrm{j}2\pi (f-f_0)}) - \mathsf{x}(e^{\mathrm{j}2\pi (f+f_0)})\right]$ |
| Parseval's Theorem | $\sum_{n=-\infty}^{\infty} \lvert x[n]\rvert^2 = \int_{-1/2}^{1/2} \left\lvert\mathsf{x}(e^{\mathrm{j}2\pi f})\right\rvert^2 \mathrm{d}f$ | |

| Function Name | Sequence | Fourier Transform |
|---|---|---|
| Impulse | $\delta[n]$ | $1$ |
| | $\delta[n-n_0]$ | $e^{-\mathrm{j}2\pi n_0 f}$ |
| DC | $1\ (-\infty < n < \infty)$ | $\sum_{k=-\infty}^{\infty} \delta\left(f-k\right)$ |
| Complex Exponential | $e^{\mathrm{j}2\pi f_0 n}$ | $\sum_{k=-\infty}^{\infty} \delta\left(f-f_0-k\right)$ |
| Cosine | $\cos(2\pi f_0 n+\theta)$ | $\sum_{k=-\infty}^{\infty} \frac{1}{2}\left[e^{\mathrm{j}\theta}\delta(f-f_0-k)\right.$ $\left.+e^{-\mathrm{j}\theta}\delta(f+f_0-k)\right]$ |
| Sine | $\sin(2\pi f_0 n+\theta)$ | $\sum_{k=-\infty}^{\infty} \frac{1}{2\mathrm{j}}\left[e^{\mathrm{j}\theta}\delta(f-f_0-k)\right.$ $\left.-e^{-\mathrm{j}\theta}\delta(f+f_0-k)\right]$ |
| Unit Step | $u[n]$ | $\frac{1}{1-e^{-\mathrm{j}2\pi f}}+\frac{1}{2}\sum_{k=-\infty}^{\infty}\delta\left(f-k\right)$ |
| | $a^n u[n]$ | $\frac{1}{1-ae^{-\mathrm{j}2\pi f}}$ |
| | $(n+1)a^n u[n]$ | $\frac{1}{(1-ae^{-\mathrm{j}2\pi f})^2}$ |
| Window | $\mathrm{rect}\left(\frac{n}{M}\right)=\begin{cases}1 & \lvert n\rvert \leqslant M\\ 0 & \lvert n\rvert > M\end{cases}$ | $\frac{\sin\left(2\pi(n+\frac{1}{2})f\right)}{\sin(\pi f)}$ |
| Sinc | $2f_{\mathrm{c}}\mathrm{sinc}\left(2f_{\mathrm{c}}n\right)$ | $\sum_{k=-\infty}^{\infty}\mathrm{rect}\left(\frac{f-k}{2f_{\mathrm{c}}}\right)$ |
| | $\frac{r^n\sin\left[2\pi f_{\mathrm{p}}(n+1)\right]}{\sin 2\pi f_{\mathrm{p}}}$ with $\lvert f_{\mathrm{p}}\rvert<1$ | $\frac{1}{1-2r\cos 2\pi f_{\mathrm{p}}e^{-\mathrm{j}2\pi f}+r^2e^{-\mathrm{j}4\pi f}}$ |

note the sum which gives periodicity

Further review of DFT provided in subsequent lectures when OFDM is introduced

# Discrete Fourier transform (DFT)




$$\text{Analysis} \quad x[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn} \quad k = 0, 1, \ldots, N-1$$

$$\text{Synthesis} \quad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} x[k] e^{j\frac{2\pi}{N}kn} \quad n = 0, 1, \ldots, N-1.$$

- DTFT is difficult to employ in real-time systems due to the continuous frequency variable
- Alternative is the DFT and its efficient implementation Fast Fourier Transform (FFT)
- Only applies to **finite length** signals

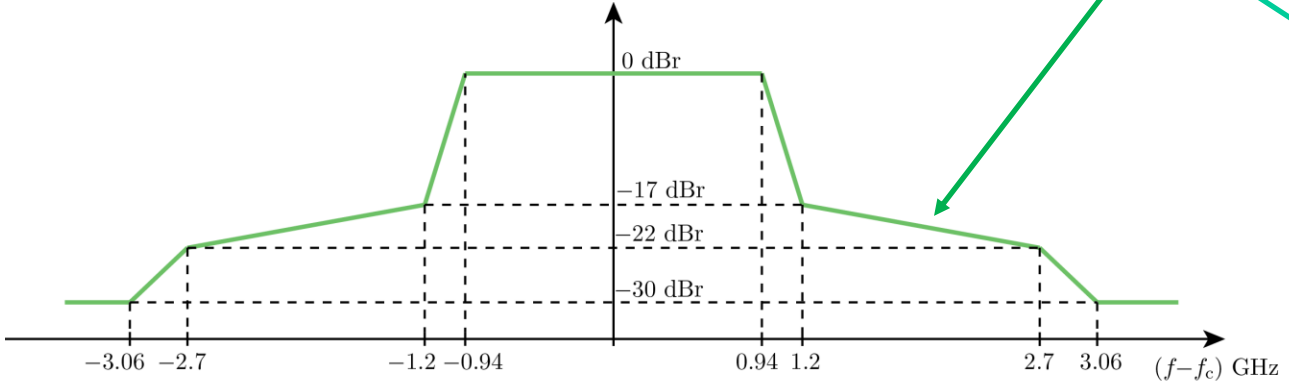
# Band limited signals in communications

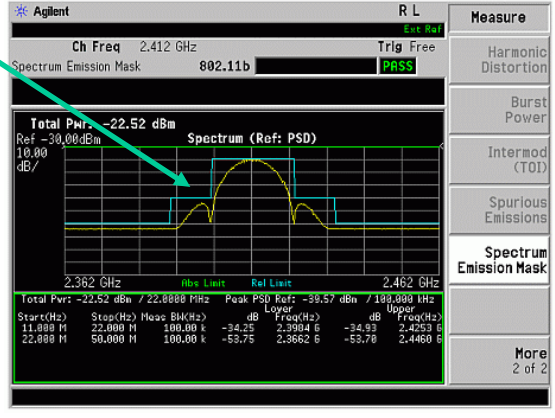
Learning objective:

- Explain the connection between communication and DSP

# Transmitted signal in wireless is bandlimited

Spectrum emission masks



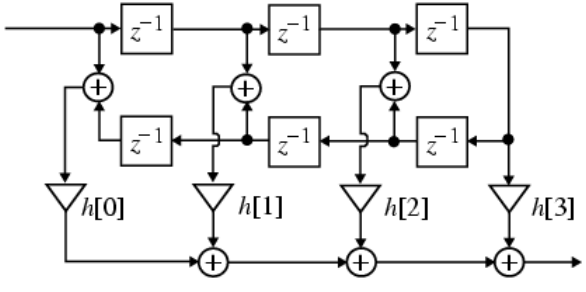




Spectrum mask for IEEE 802.11ad

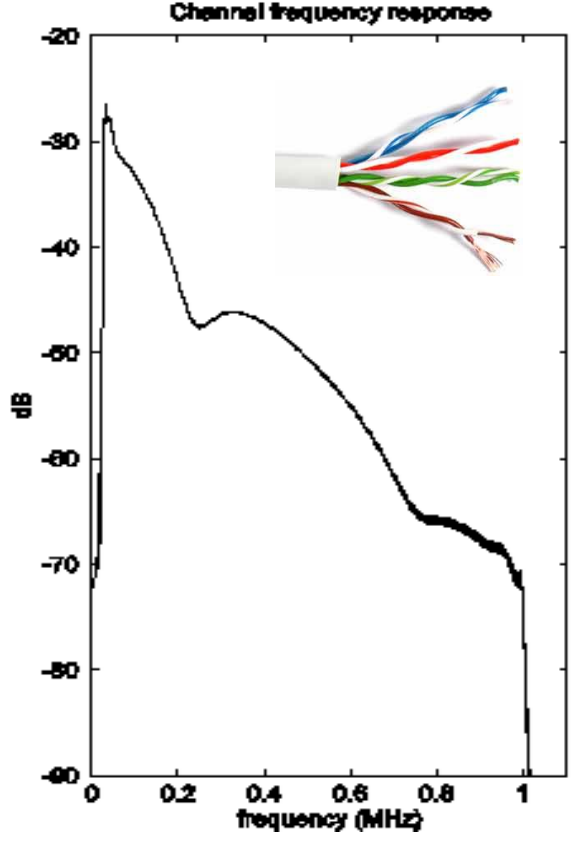
- It can not extend much beyond the allocated bandwidth
- Bandlimited signals can be represented in discrete-time!
  - Use DSP mathematical tools to design and analyze bandlimited signals

# Only care about bandlimited part of the channel

- The degree to which different frequencies are amplified or attenuated by the channel is different
- Bandlimited channels also represented in discrete-time!
  - ✦ Use DSP mathematical tools to design and analyze bandlimited signals

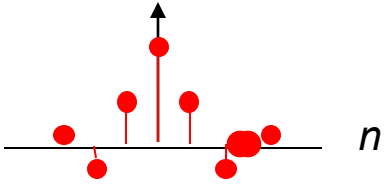



**DSL channel**
**Band selector filter (30 kHz - 1 MHz)**



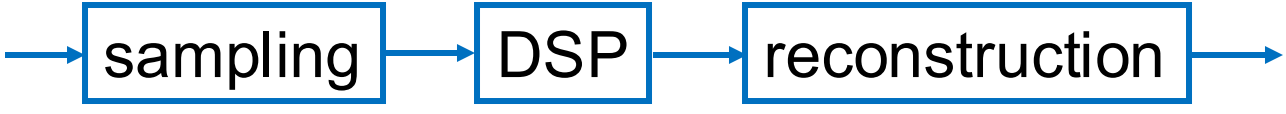
# Received signal is band limited

- Use Nyquist's Sampling Theorem to represent signals in discrete-time




sampling

- Design efficient algorithms to process bandlimited signals




The bandlimited property makes the connection between comm. and DSP

# Band limited signals in communications

Learning objective:

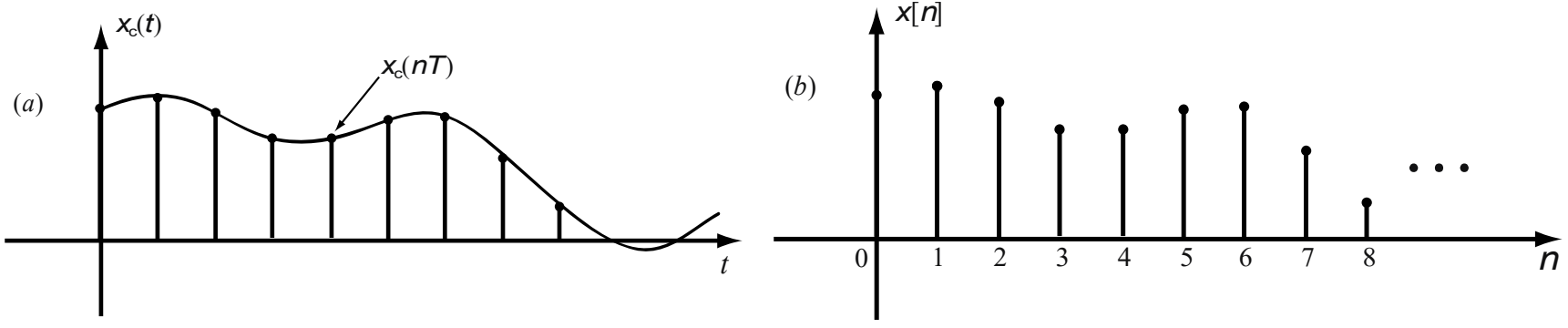
- Explain the connection between communication and DSP

# Sampling

Learning objectives:

- Determine the Nyquist sampling frequency for a given signal.
- Explain the theory behind sampling and reconstruction.
- Demonstrate aliasing.

# Nyquist sampling theorem




- Let $x(t)$ be a bandlimited signal, which means that $X(f) = 0 \, for \, f \geqslant f_N$
- $x(t)$ is uniquely determined by its samples $\{x[n] = x(nT)\}_{n=-\infty}^{\infty}$ if

$$f_s = \frac{1}{T} > 2f_N$$

where $f_N$ is the Nyquist frequency and $2f_N$ is generally known as the **Nyquist rate**

# Reconstruction with the sampling theorem

- $x(t)$ can be reconstructed from its samples

$$x(t) = \sum_n x[n]\text{sinc}\left(\frac{t - nT}{T}\right)$$

- Relation between the spectra of discrete time and continuous time signals

$$x\left(e^{j2\pi f}\right) = \frac{1}{T}\sum_n x\left(\frac{f}{T} - \frac{n}{T}\right)$$

- The spectra of the discrete time signal is periodic as expected
- Why do we have the scaling factor 1/T in the equation?

# Example of aliasing

$$x\left(e^{j2\pi f}\right) = \frac{1}{T}\sum_{n} x\left(\frac{f}{T} - \frac{n}{T}\right)$$

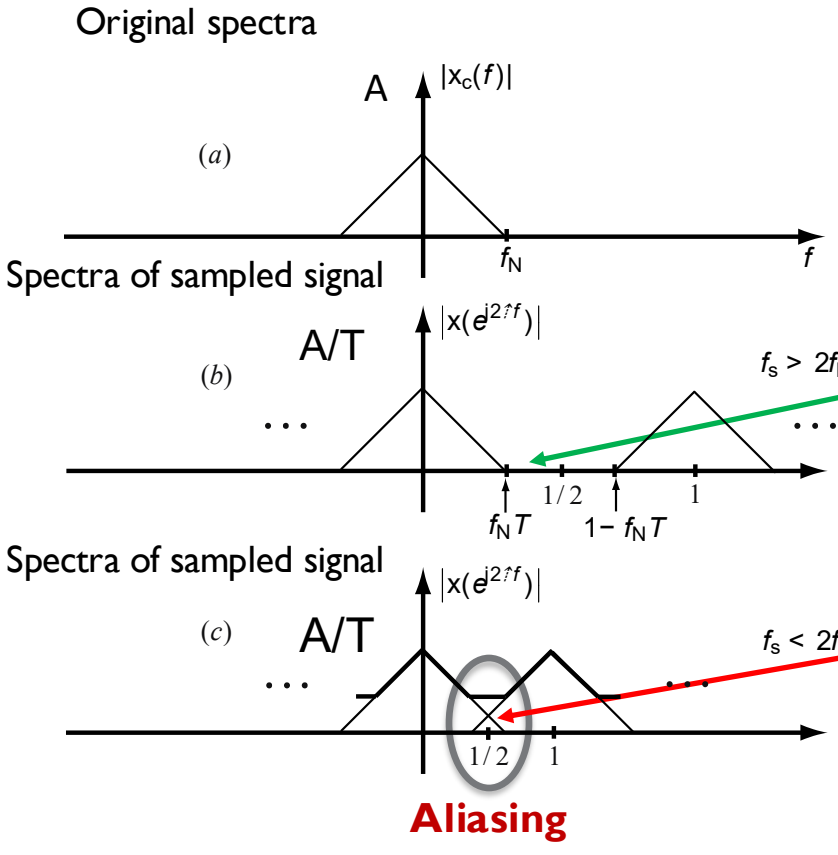



figure out how the triangle scales

$$\frac{f}{T} = f_N$$

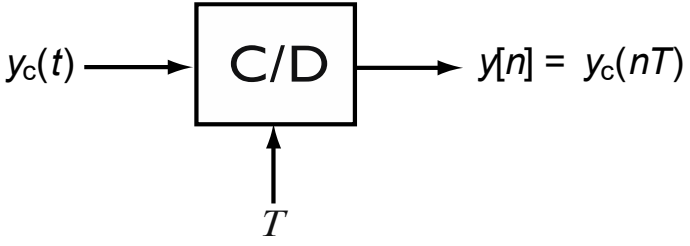
$$\rightarrow \quad f = T f_N$$

Nyquist criterion is satisfied

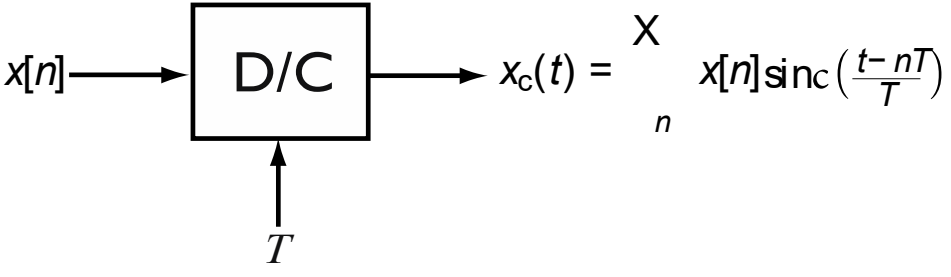
Nyquist criterion is not satisfied

# Sampling notation

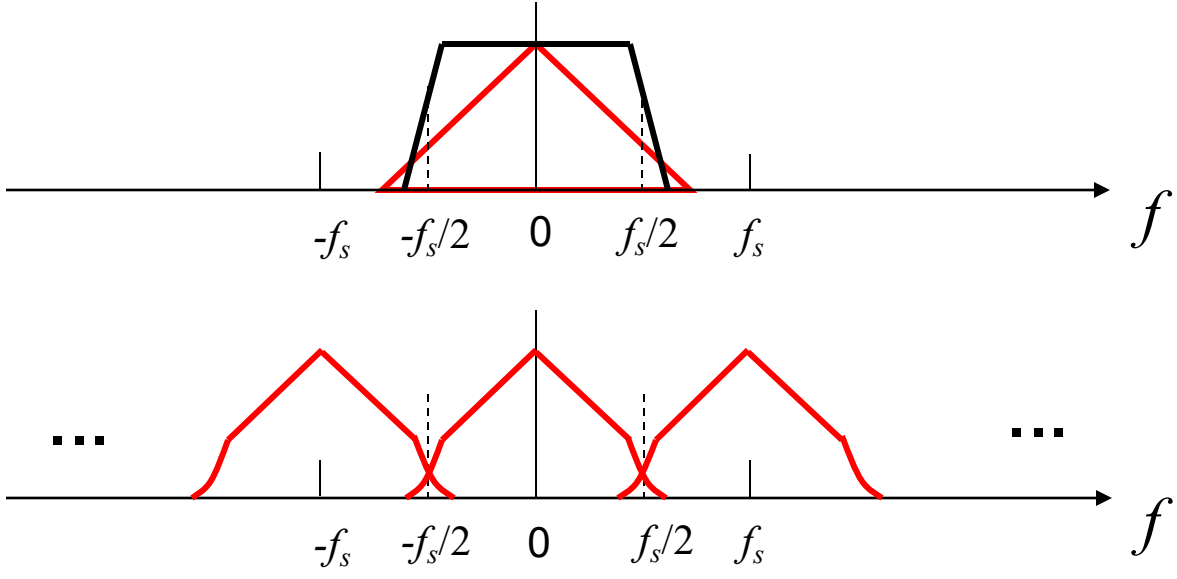
- C/D: continuous-to-discrete converter
  - Ideal analog-to-digital converter




- D/C: discrete-to-continuous converter
  - Ideal digital-to-analog converter

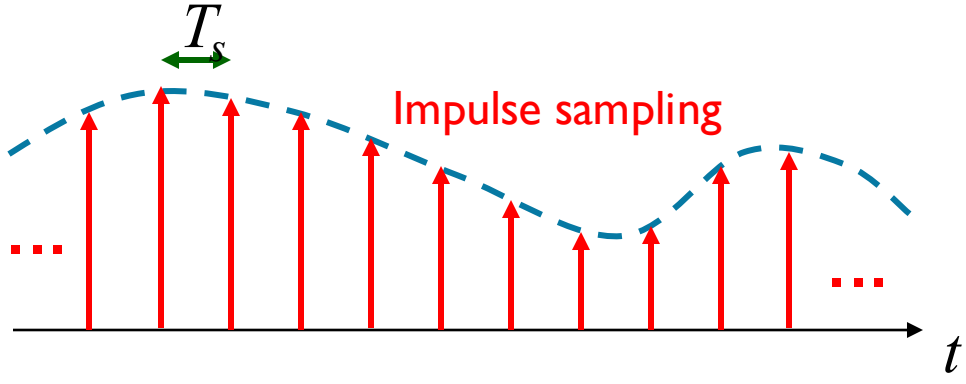


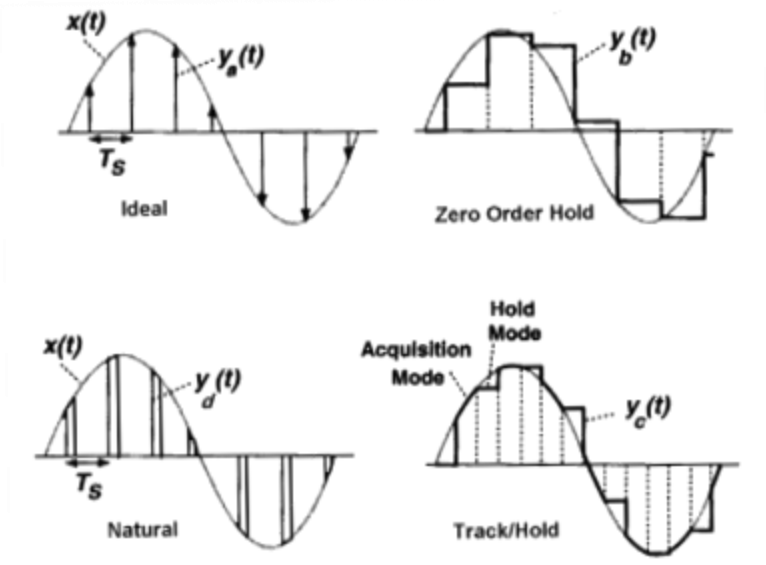
# Antialiasing filter




- Use an analog antialiasing filter (AAF) before sampling
- Better to filter out high frequencies than to have aliasing distortion
- AAF also avoids out-of-band noise aliasing

# Sampling techniques




- **Impulse Sampling**: theoretical – not implemented in practice
- **Natural Sampling**: theoretical - multiplier is a switch
- **Zero-order hold Sampling:** Ideal Sample/Hold – instantaneous acquisition time is impractical
- **Track/Hold:** Result is sampled and stored in a memory element

# Sampling

Learning objectives:

- Determine the Nyquist sampling frequency for a given signal.
- Explain the theory behind sampling and reconstruction.
- Demonstrate aliasing.